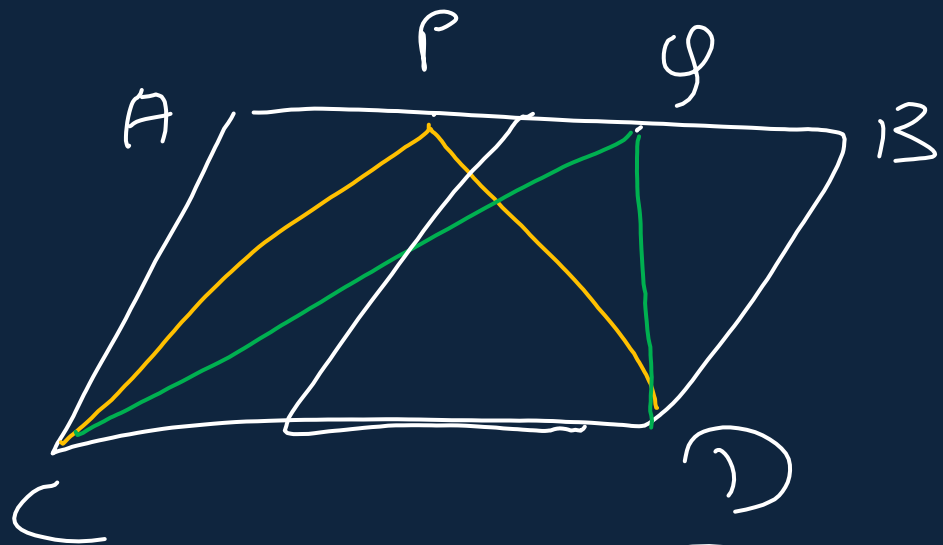




$$\triangle PCQ =$$

$$\begin{aligned} \text{Base} &= CD \\ \text{height} &= h \end{aligned}$$

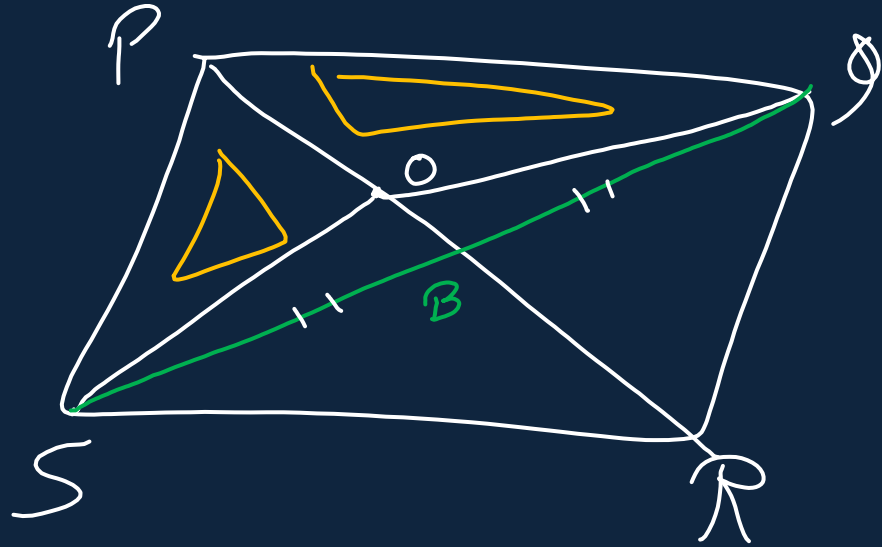
$$\triangle QCD =$$

$$\begin{aligned} \text{Base} &= CD \\ \text{height} &= h \end{aligned}$$

$$10 \times \frac{\frac{1}{2} \times 6 \times h}{\frac{1}{2} \times 6 \times h}$$

$$= \frac{1}{1}$$





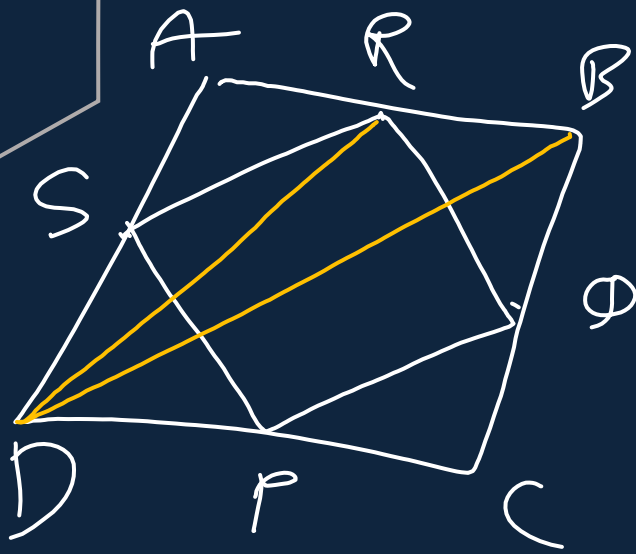
$$\text{Area } \triangle PBS = \text{Area } \triangle PBQ \quad \{ \}$$

$$\triangle OBS = \triangle OBQ$$

$$|\text{Area}(PBS - OBS) = \text{Area}(PBQ - OBQ)|$$

$$\triangle POB = \triangle POQ$$

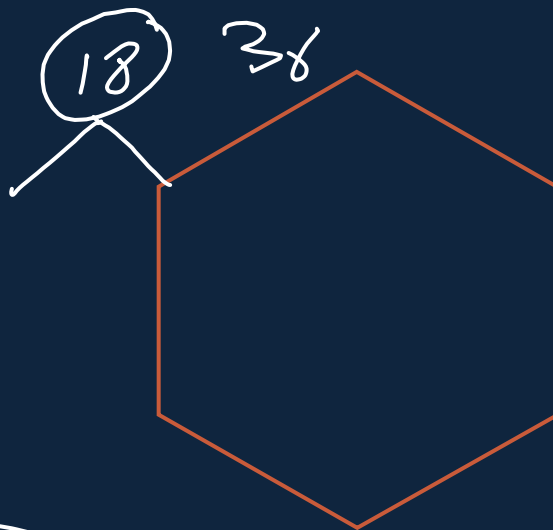




$$\begin{aligned} \text{Area } \triangle ARD &= \frac{1}{2} \text{ Area of } \triangle ABD \\ \text{Area } \triangle SRD &= \frac{1}{2} \text{ Area of } \triangle ARD \\ \text{Area } \triangle SRD &= \frac{1}{2} \left(\frac{1}{2} \text{ Area of } \triangle ABD \right) \end{aligned}$$



$$\overbrace{2x^2 + 9x + 18}^{\times}$$



4

$$x^2 + 9x$$

$$\overbrace{44x^2 - x - 3}^{\times}$$

$$44x^2 + 11x - 12x - 3$$

$$11x(4x + 1)$$

$$3(4x + 1)$$

$$(4x + 1)(11x - 3)$$

$$\begin{array}{r} 132 \\ +11 \quad -12 \end{array}$$

$$-1$$

$$2x^3 + 432m^3$$

$$2(\underline{x^3} + \underline{216m^3})$$



$$1^2 - 24k = 0$$

$$1 = 24k$$

$$k = \frac{1}{24}$$

A.P.

$$\boxed{366} = n$$

(4) 7 10 13 16

$$\cancel{2-4} = \cancel{3}$$

4 7 9 11 10
f

$$4 \quad t_n = \frac{a + (n-1)d}{2}$$

a = first term

d = difference

n = ~~the~~ number of terms



$$\frac{2 \times 3}{2 \times 1} = \frac{5}{2}$$

$$\frac{8}{2} = \frac{5}{2}$$

$$= \left(\frac{1}{2} \right)$$

$$= \left(\frac{1}{2} \right)$$

$$\frac{7}{3} = \frac{3 \times 3}{1 \times 3}$$

$$\frac{7}{3} = \frac{9}{3}$$

$$= \left(\frac{2}{3} \right)$$

$$2, \frac{5}{2}, 3, \frac{7}{2}$$

$$\frac{5}{2} - \frac{2 \times 2}{1 \times 2} = \frac{5}{2} - \frac{4}{2}$$

$$= \left(\frac{1}{2} \right)$$

$$\sin^4 \theta - \cos^4 \theta = 1 - 2\cos^2 \theta$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$\cos = (\sin^2 \theta)^2 - (\cos^2 \theta)^2$$

$$= (\sin^2 \theta + \cos^2 \theta) (\sin^2 \theta - \cos^2 \theta)$$

$$= 1 \cdot (\sin^2 \theta - \cos^2 \theta)$$

$$1 - \cos^2 \theta - \cos^2 \theta$$

$$1 - 2\cos^2 \theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$-6 - (-10)$$

$$-6 + 10$$

$$\begin{aligned} & \textcircled{4} \\ & -2 - (-8) \\ & -2 + 8 \end{aligned}$$

$$\begin{aligned} & \textcircled{4} \\ & 2 - (-2) = 2 + 2 = \textcircled{4} \end{aligned}$$

$$5(x+2) + 10y(x+2)$$

$$5(x+2)(1+2y)$$



$$\begin{array}{r} 90 \\ - 04 \\ \hline \end{array} \quad 00400$$

$$\begin{array}{r} 0.333 \\ - 0.330 \\ \hline \end{array}$$

$$0.003$$

$$\begin{array}{r} 0.33 \\ - 0.30 \\ \hline \end{array}$$

$$0.03$$

$$d = -40$$

$$S_n = 1360 + 8000 = 9360$$

$$n = 12$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$9360 = \frac{12}{2} [2a + 11 \times (-40)]$$

$$1560 = 2a - 440$$

$$1560 = 2a - 440$$

$$1560 + 440 = 2a$$

$$2000 = 2a$$

$$a = 1000$$

$$a + 11d = 1000 + 11 \times (-40) = 1000 - 440$$

$$\begin{array}{r} 1560 \\ 440 \\ \hline 2000 \end{array}$$

$$\tan \theta + \frac{1}{\tan \theta} = 2$$

$$\tan^2 \theta + \frac{1}{\tan^2 \theta} + 2 \tan \theta \times \frac{1}{\tan \theta} = 4$$

$$= 4 - 2^{11} = 2$$

$$\frac{12}{2} [2a + 11 \times (-40)] = 9360$$